Implementation of Gaussian processes regression

- Numerical implementation
- Non-standard kernels

GP regression

We have data $\left\{\left(x_n^\sharp, y_n^\sharp, \epsilon_n^\sharp\right)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R} \times \mathbb{R}$ and seek an unknown function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

- Our regression model consists of

$$\begin{aligned} f &\sim \mathrm{GP}_{\mathbb{X}}\left(m, v\right) \\ y_n^\sharp | f &\sim \mathrm{Normal}\left(f(x_n^\sharp), \upsilon_n^\sharp\right), \qquad n = 1, \ldots, N^\sharp \end{aligned}$$

- To obtain predictions at some $x^\star \in \mathbb{X}$, we apply the predictive distribution

$$\begin{aligned} f^\star | Y^\sharp &\sim \mathrm{Normal}_1\left(\mu^\star, \upsilon^\star\right) && f^\star \equiv f(x^\star) \\ \mu^\star &= m^\star + V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}\left(Y^\sharp - m^\sharp\right) && \mu^\star \equiv \mu(x^\star) \\ \upsilon^\star &= V^{\star\star} - V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1} V^{\sharp\star} && \upsilon^\star \equiv \upsilon(x^\star) \end{aligned}$$

Implementation of the predictive statistics

Our predictive statistics, at input $x^\star \in \mathbb{X}$, are determined by

$$\mu^\star = m^\star + V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}\left(Y^\sharp - m^\sharp\right) \qquad \mu^\star \equiv \mu(x^\star)$$

$$\upsilon^\star = V^{\star\star} - V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1} V^{\sharp\star} \qquad \upsilon^\star \equiv \upsilon(x^\star)$$

Evaluation of μ^*, υ^* starts with the (lower) Cholesky factor *...which is computed in time* $O\left((N^\sharp)^3\right)$

$$\left(L^{\sharp\sharp}\right)\left(L^{\sharp\sharp}\right)^t = P^{\sharp\sharp} + V^{\sharp\sharp}$$

and proceeds via the vector *...which is computed in time* $O\left((N^\sharp)^2\right)$

$$\alpha^\sharp = \left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}\left(Y^\sharp - m^\sharp\right) \iff \left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)\alpha^\sharp = Y^\sharp - m^\sharp \iff \left(L^{\sharp\sharp}\right)\left(L^{\sharp\sharp}\right)^t\alpha^\sharp = Y^\sharp - m^\sharp$$

and the matrix

$$V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1} V^{\sharp\star} = V^{\star\sharp}\left((L^{\sharp\sharp})(L^{\sharp\sharp})^t\right)^{-1} V^{\sharp\star} = V^{\star\sharp}(L^{\sharp\sharp})^{-t}(L^{\sharp\sharp})^{-1}V^{\sharp\star} = \left((L^{\sharp\sharp})^{-1}V^{\sharp\star}\right)^t\left((L^{\sharp\sharp})^{-1}V^{\sharp\star}\right)$$

...given $\alpha^\sharp$, *the predictive mean* $\mu^\star$ *is obtained in time* $O(N^\sharp)$
...given $L^{\sharp\sharp}$, *the predictive variance* $\upsilon^\star$ *is obtained in time* $O\left((N^\sharp)^2\right)$

Implementation of the Cholesky factor

We consider the (lower) Cholesky factor

$$\left(L^{\sharp\sharp}\right)\left(L^{\sharp\sharp}\right)^{t} = P^{\sharp\sharp} + V^{\sharp\sharp}$$

Our $L^{\sharp\sharp}$ is based on the discretization

$$x^{\sharp} = \begin{pmatrix} x_1^{\sharp} \\ \vdots \\ x_{N^{\sharp}}^{\sharp} \end{pmatrix}$$

The direct computation of $L^{\sharp\sharp}$ is achieved in time $O\left((N^{\sharp})^3\right)$

We partition $x^{\sharp}$ like this

$$x^{\sharp} = \begin{pmatrix} x_1^{\sharp} \\ \vdots \\ x_{N^{\sharp}}^{\sharp} \end{pmatrix} \equiv \begin{pmatrix} x_{old} \\ x_{add} \end{pmatrix} = x_{new}$$

and consider the corresponding Cholesky factors

$$N^{\sharp} \leftrightarrow x^{\sharp} \leftrightarrow L^{\sharp\sharp} \equiv L_{new} \leftrightarrow x_{new} \leftrightarrow N_{new}$$
$$L_{old} \leftrightarrow x_{old} \leftrightarrow N_{old}$$

According to this scheme, L_{new} can be obtained sequentially

Implementation of the Cholesky factor

Our partitioned discretization and sequential discretization scheme is

$$x_{new} = \begin{pmatrix} x_{old} \\ x_{add} \end{pmatrix} \qquad L_{old} \leftrightarrow x_{old} \leftrightarrow N_{old} \qquad L_{new} \leftrightarrow x_{new} \leftrightarrow N_{new} \qquad N_{new} = N_{old} + N_{add}$$

The (lower) Cholesky factors are defined by the decompositions

$$L_{old} L_{old}^t = P_{old} + V_{old} \qquad L_{new} L_{new}^t = P_{new} + V_{new}$$

and are obtained by

$$P_{new} = \begin{bmatrix} P_{old} & 0 \\ 0 & P_{add} \end{bmatrix} \qquad V_{new} = \begin{bmatrix} V_{old} & v_{old,add} \\ v_{add,old} & V_{add} \end{bmatrix}$$

Accordingly, the updated Cholesky factor is

$$L_{new} L_{new}^t = \begin{bmatrix} P_{old} + V_{old} & v_{old,add} \\ v_{add,old} & P_{add} + V_{add} \end{bmatrix}$$

This indicates that

$$L_{new} = \begin{pmatrix} L_{old} & 0 \\ \Lambda_1 & \Lambda_2 \end{pmatrix}, \qquad \Lambda_2 \Lambda_2^t = P_{add} + V_{add} - \Lambda_1 \Lambda_1^t, \qquad \Lambda_1 = v_{add,old} L_{old}^{-t}$$

...this way the computation of L_{new} requires only time $O(N_{add} N_{old})$

The squared exponential kernel

The square exponential kernel is

$$v(x', x'') = \sigma^2 \exp\left(-\frac{1}{2}\left(\frac{x' - x''}{\ell}\right)^2\right) \qquad x', x'' \in \mathbb{R}$$

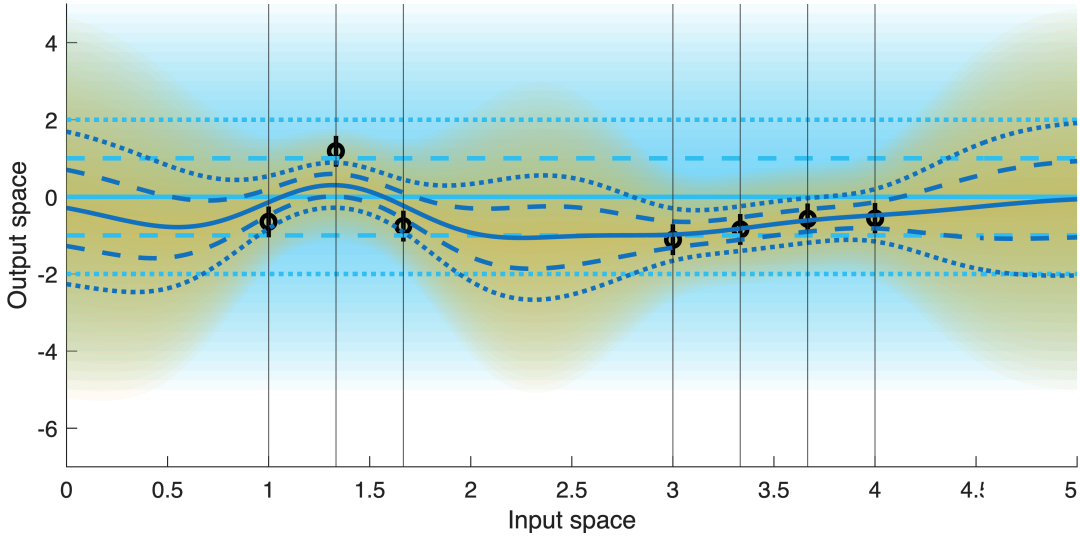
The kernel parameters consist of a strength σ and a scale ℓ

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2\left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix}\right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \exp\left(-\frac{1}{2}\left(\frac{x' - x''}{\ell}\right)^2\right)$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
5

The white noise kernel

The white noise kernel is

$$v(x', x'') = \sigma^2 \cdot \begin{cases} 1, & x' = x'' \\ 0, & x' \neq x'' \end{cases} \qquad x', x'' \in \mathbb{R}$$

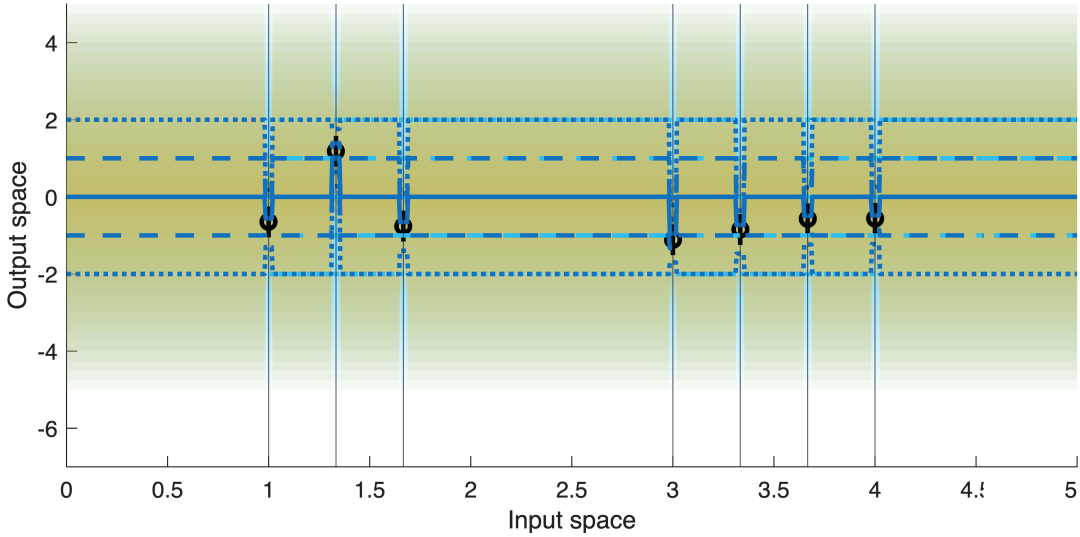
The only kernel parameter consist of a strength σ

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2 \left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix} \right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \begin{cases} 1, & x' = x'' \\ 0, & x' \neq x'' \end{cases}$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

The constant kernel

The constant kernel is

$$v(x', x'') = \sigma^2 \qquad\qquad x', x'' \in \mathbb{R}$$

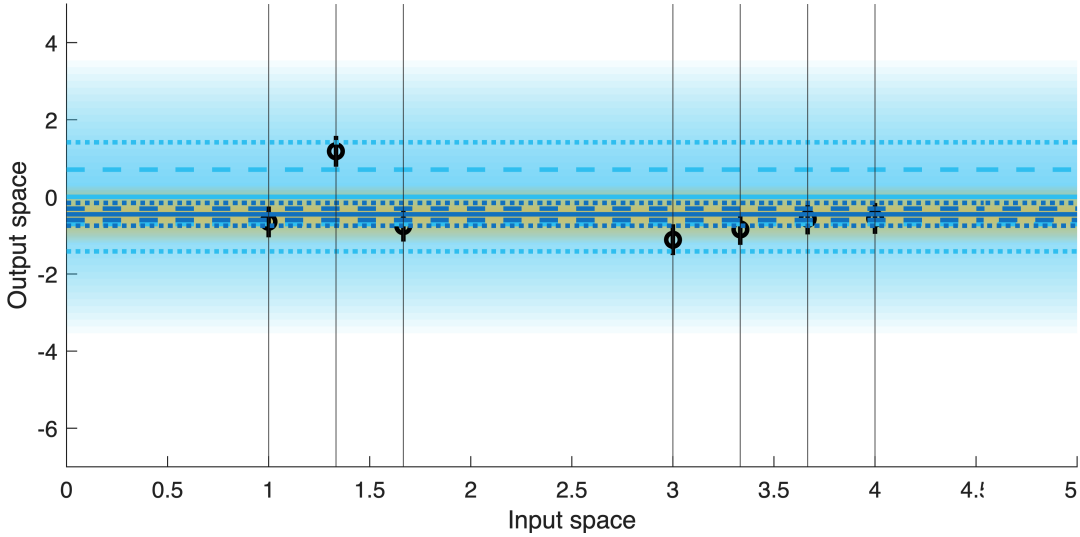
The only kernel parameter consist of a strength σ

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2 \left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix} \right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = 1$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

The exponential sine squared kernel

The constant kernel is

$$v(x', x'') = \sigma^2 \exp\left(-\frac{2}{\ell^2}\sin^2\left(\frac{\pi}{p}(x' - x'')\right)\right) \qquad x', x'' \in \mathbb{R}$$

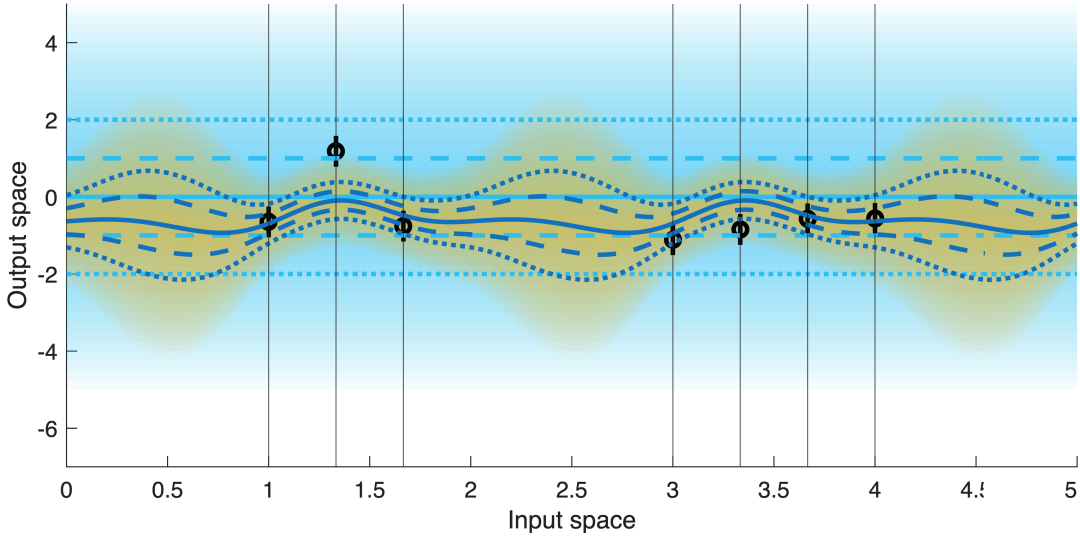
The kernel parameters consist of a strength σ, a scale ℓ, and a period p

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2\left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix}\right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \exp\left(-\frac{2}{\ell^2}\sin^2\left(\frac{\pi}{p}(x' - x'')\right)\right)$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
5

The diffusion kernel

The diffusion kernel is

$$v(x', x'') = 2D \min\{|x'|, |x''|\} \qquad x', x'' \in \mathbb{R}$$

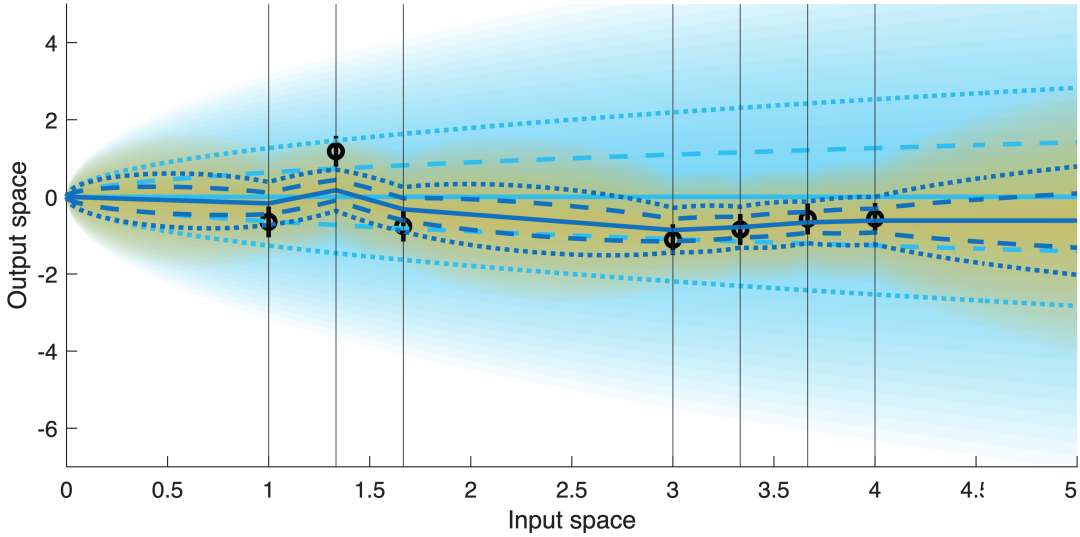
The only kernel parameter consists of a diffusivity D

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2 \left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix} \right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \frac{\min\{|x'|, |x''|\}}{\sqrt{|x' x''|}}$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
5

The anchored diffusion kernel

The anchored diffusion kernel is

$$v(x', x'') = 2D \cdot \begin{cases} \min\{|x' - X|, |x'' - X|\}, & x' > X, x'' > X \\ \min\{|x' - X|, |x'' - X|\}, & x' < X, x'' < X \\ 0, & x' > X, x'' < X \\ 0, & x' < X, x'' > X \end{cases} \qquad x', x'' \in \mathbb{R}$$

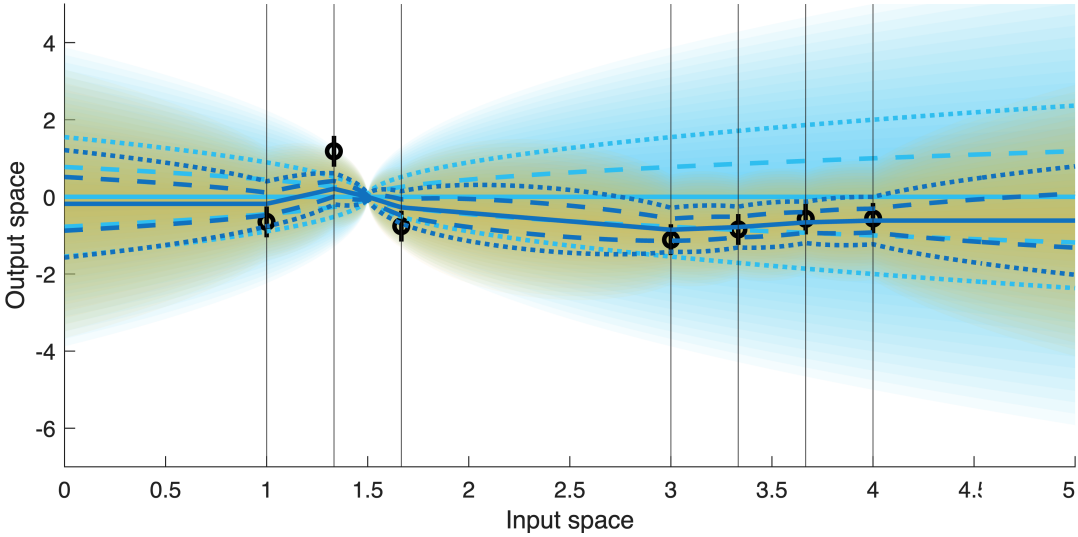
The kernel parameters consist of a diffusivity D and an anchor X

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \text{Normal}_2 \left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix} \right)$$

Our correlation is given by

$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \begin{cases} \frac{\min\{|x' - X|, |x'' - X|\}}{\sqrt{(x' - X)(x'' - X)}}, & x' > X, x'' > X \\ \frac{\min\{|x' - X|, |x'' - X|\}}{\sqrt{(x' - X)(x'' - X)}}, & x' < X, x'' < X \\ 0, & x' > X, x'' < X \\ 0, & x' < X, x'' > X \end{cases}$$


Output space
Input space
4
2
0
-2
-4
-6
0
0.5
1
1.5
2
2.5
3
3.5
4
5